AWS VPC Peering Project Report

Project Title: AWS VPC Peering – Connecting Two Private Networks

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Batch: 1
AWS Region: Asia Pacific (Sydney) – ap-southeast-2
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1. Introduction

The objective of this project is to establish private communication between two Amazon Virtual Private Clouds (VPCs) using AWS VPC Peering.

The project uses two VPCs with non-overlapping IPv4 CIDR blocks. Each VPC contains an application server, and a bastion host in VPC A provides secure SSH administration access to both application servers.

The project demonstrates VPC creation, subnet configuration, route table management, security group configuration, SSH connectivity, and communication testing using private IP addresses.

2. Project Objectives

- Create two VPCs with non-overlapping CIDR blocks.
- Configure public and private subnets with appropriate route tables.
- Launch one bastion host and two private EC2 instances.
- Establish and activate a VPC peering connection.
- Configure routes in both VPCs and the bastion's public subnet.
- Enable controlled SSH and ICMP traffic using security groups.
- Verify connectivity before and after peering.
- Perform a controlled connectivity failure and recovery test.

3. Architecture Overview

The project consists of two VPCs connected through a VPC peering connection.

VPC A

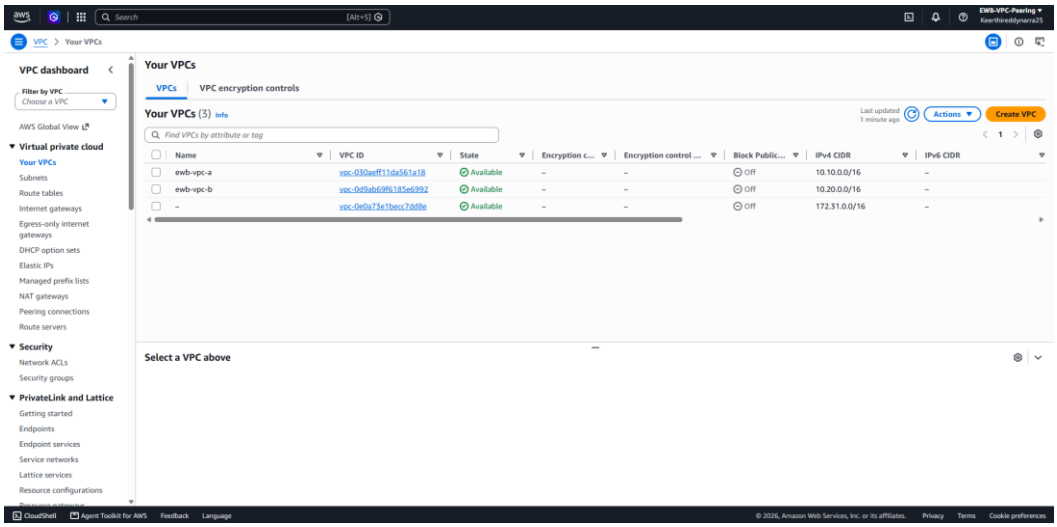
- CIDR: 10.10.0.0/16
- Public subnet: 10.10.1.0/24
- Private subnet: 10.10.2.0/24
- Bastion host: 10.10.1.10
- App-A: 10.10.2.10

VPC B

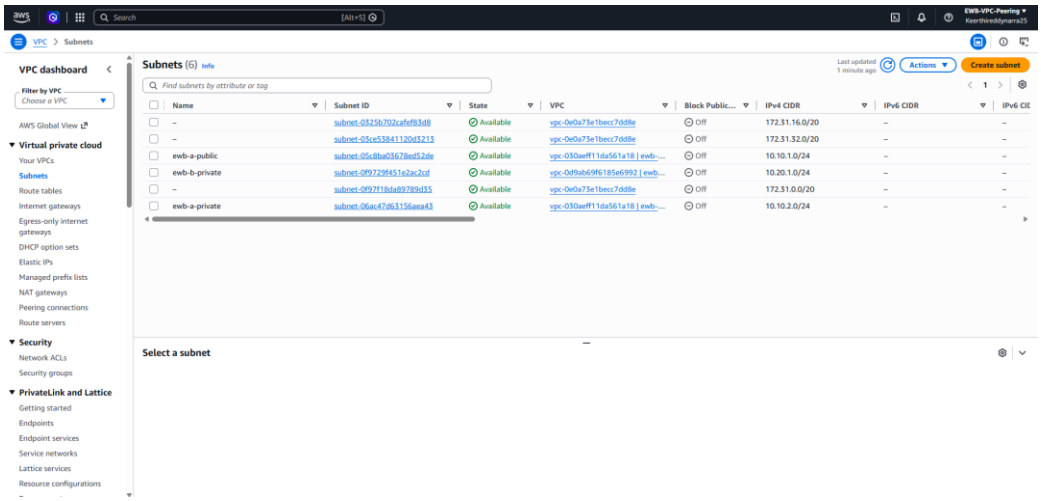
- CIDR: 10.20.0.0/16

- Private subnet: 10.20.1.0/24
- App-B: 10.20.1.10

The bastion host provides the administrative SSH path from the local computer to the private application servers. App-A and App-B communicate directly using their private IP addresses through the VPC peering connection.



VPC_Overview



Subnet_Overview

4. AWS Resource Inventory

Resource	Name	Resource ID / Address
VPC A	ewb-vpc-a	vpc-030aeff11da561a18
VPC B	ewb-vpc-b	vpc-0d9ab69f6185e6992

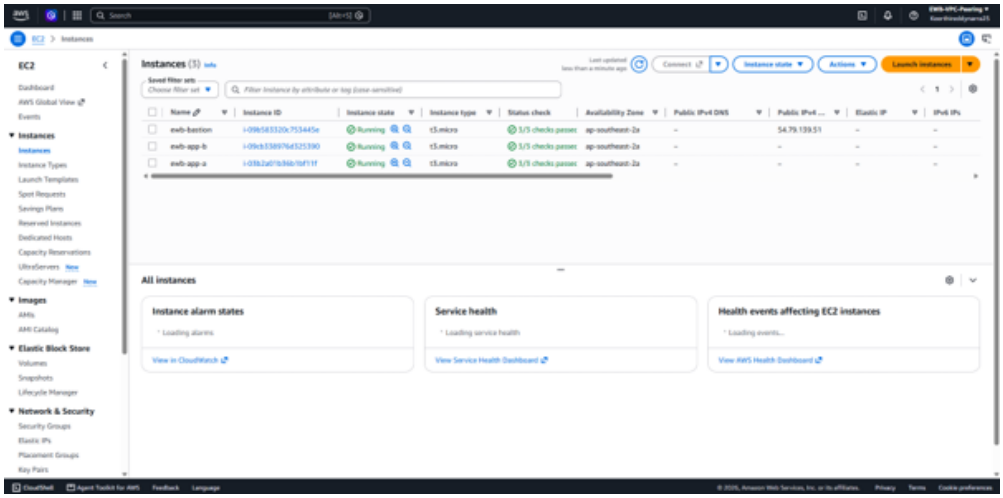
Public subnet A	ewb-a-public	subnet-05c8ba03678ed52de
Private subnet A	ewb-a-private	subnet-06ac47d63156aea43
Private subnet B	ewb-b-private	subnet-0f9729f451e2ac2cd
Internet gateway	ewb-igw-a	igw-05456be81e68372be
Public route table	ewb-rt-a-public	rtb-06d98fbc05ad211c5
Private route table A	ewb-rt-a-private	rtb-0f69e34cf8868a21d
Private route table B	ewb-rt-b-private	rtb-050cd80fd91115329
VPC peering	ewb-vpc-peering	pcx-065e6374381912571
Bastion instance	ewb-bastion	i-09b583320c753445e
App-A instance	ewb-app-a	i-03b2a01b36b1bf11f
App-B instance	ewb-app-b	i-09cb338976d325390

5. EC2 Instance Configuration

Instance	Private IP	Public IP
Bastion	10.10.1.10	54.79.139.51*
App-A	10.10.2.10	None
App-B	10.20.1.10	None

All three instances use Amazon Linux 2023 and the ec2-user account. The ewb-lab key pair is used for SSH authentication.

Verify the bastion's current public IP before final submission, as a public IP can change if it is automatically assigned.



EC2_Instance

6. Route Table Configuration

6.1 VPC A Public Route Table

Route table: ewb-rt-a-public

Subnet association: ewb-a-public

Destination	Target	Purpose
10.10.0.0/16	Local	Communication within VPC A
0.0.0.0/0	Internet gateway	Bastion internet access
10.20.0.0/16	VPC peering	Bastion access to VPC B

6.2 VPC A Private Route Table

Route table: ewb-rt-a-private

Subnet association: ewb-a-private

Destination	Target	Purpose
10.10.0.0/16	Local	Communication within VPC A
10.20.0.0/16	VPC peering	App-A access to VPC B

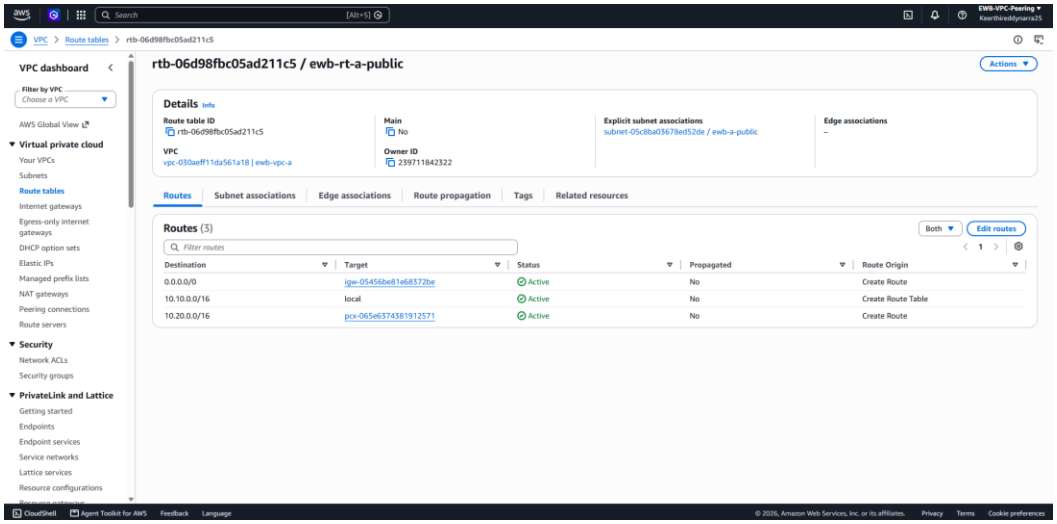
6.3 VPC B Private Route Table

Route table: ewb-rt-b-private

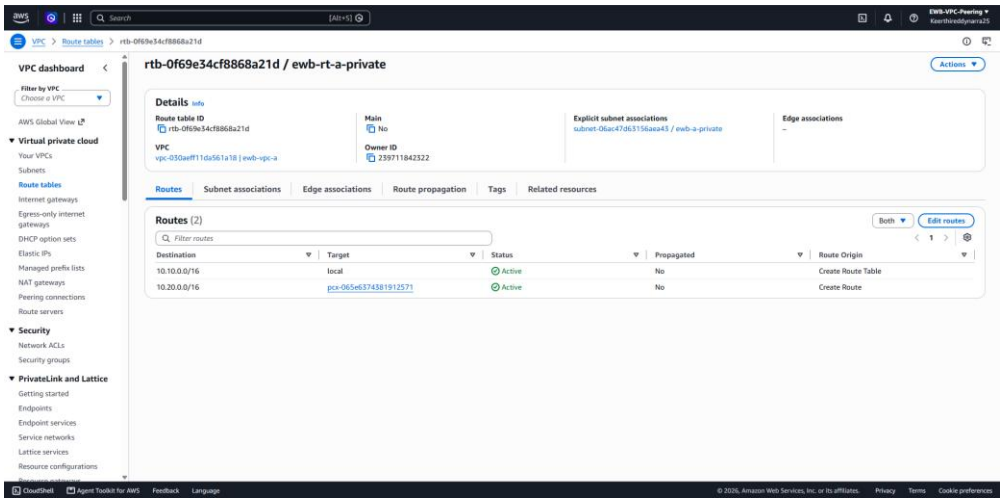
Subnet association: ewb-b-private

Destination	Target	Purpose
10.20.0.0/16	Local	Communication within VPC B
10.10.0.0/16	VPC peering	Return path to VPC A

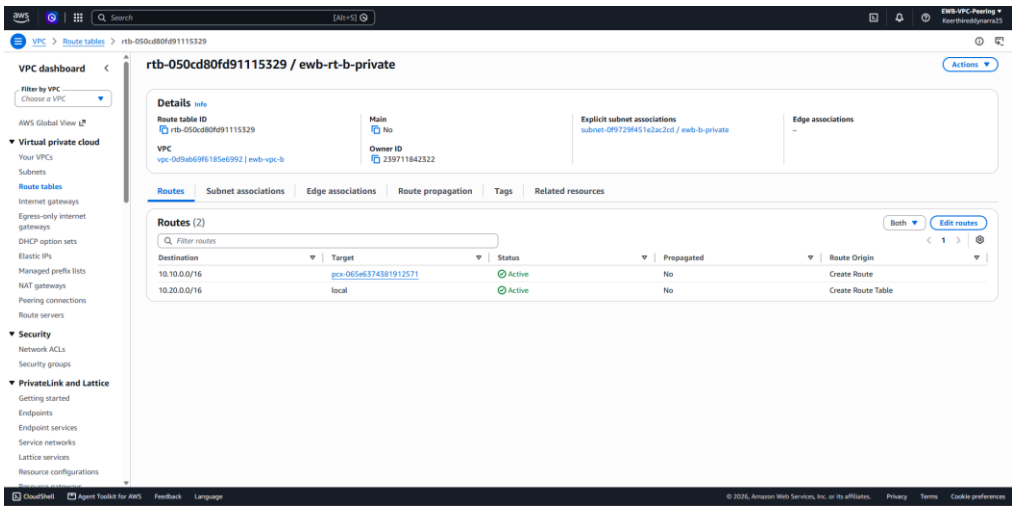
The peering routes provide forward and return paths for communication between the two VPCs.



VPC_A_Public_Route_Table



VPC_A_Private_Route_Table



VPC_B_Private_Route_Table

7. Security Group Configuration

Security groups control which traffic can reach the EC2 instances.

Bastion Security Group – ewb-sg-bastion

- Inbound SSH on TCP port 22 from the current trusted public IPv4 address /32.
- Default outbound rule allows all traffic.

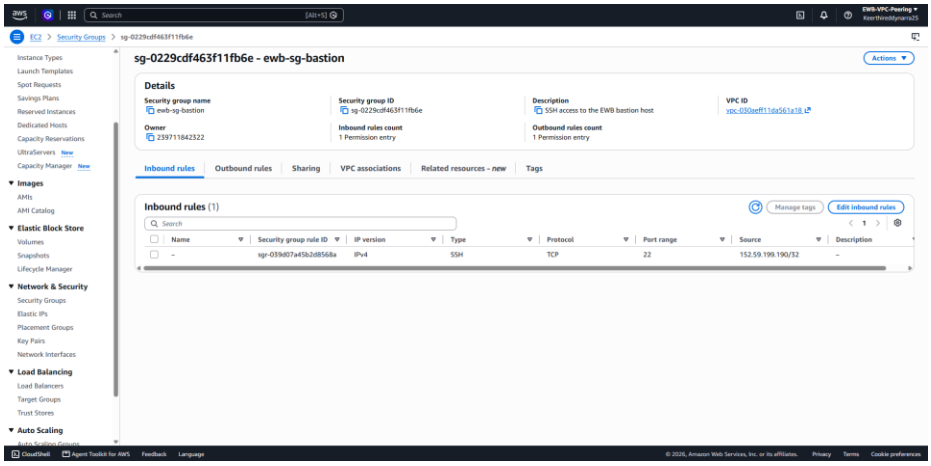
App-A Security Group – ewb-sg-app-a

- SSH on TCP port 22 from 10.10.1.10/32.
- ICMP Echo Request from 10.20.1.10/32.
- Default outbound rule allows all traffic.

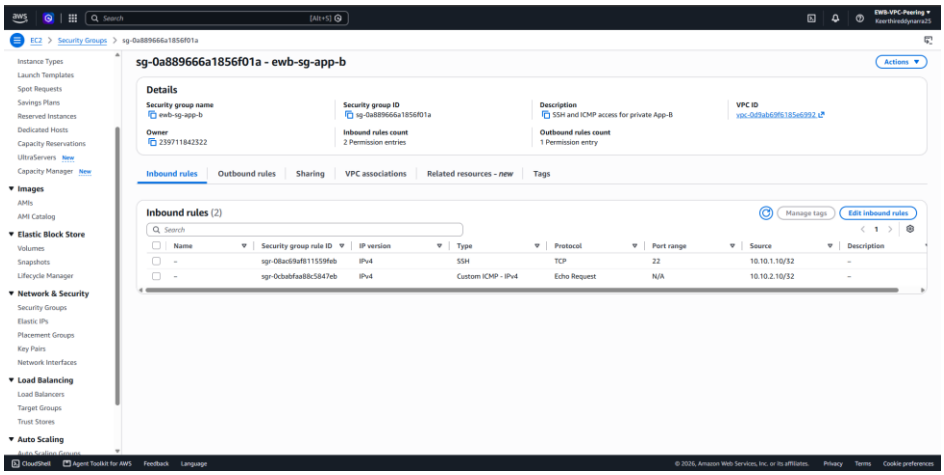
App-B Security Group – ewb-sg-app-b

- SSH on TCP port 22 from 10.10.1.10/32.
- ICMP Echo Request from 10.10.2.10/32.
- Default outbound rule allows all traffic.

The application servers have no public IP addresses. SSH access is provided through the bastion, and ICMP rules permit the intended connectivity tests.



Bastion_Security_Group



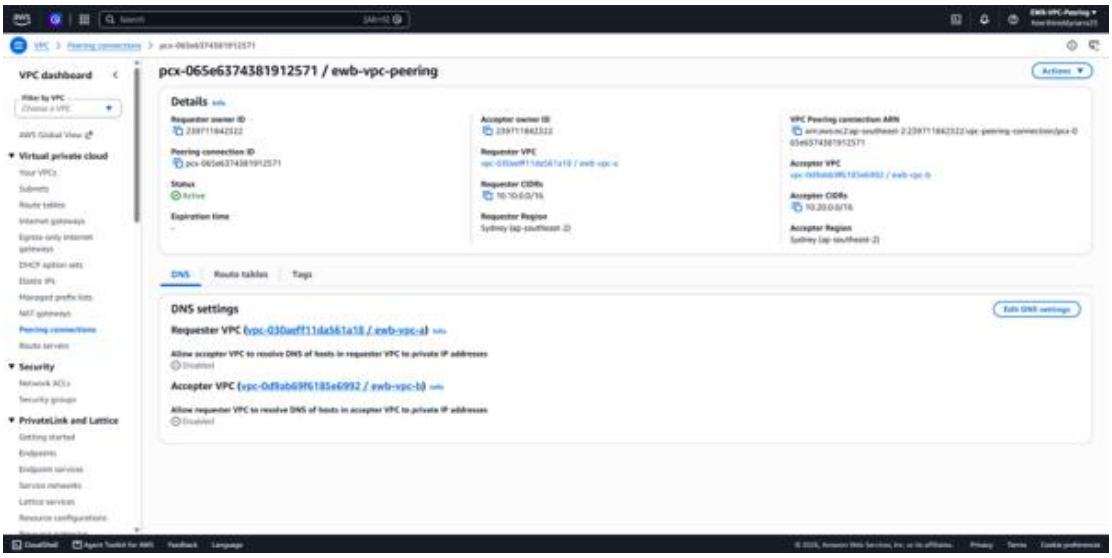
App_B_Security_Group

8. VPC Peering Configuration

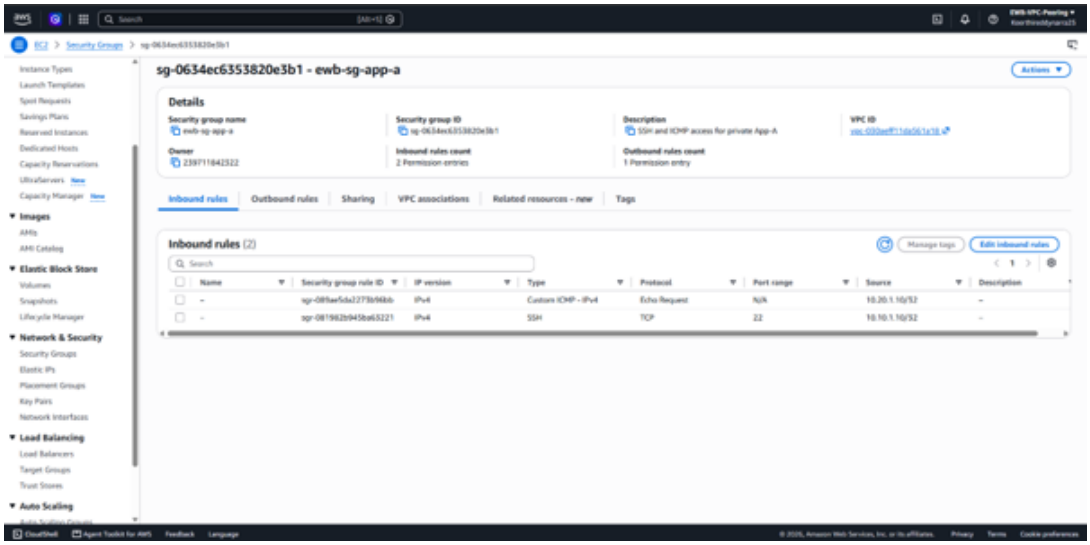
A VPC peering connection named ewb-vpc-peering was created between VPC A and VPC B in the same AWS account and Region.

- Peering connection ID: pcx-065e6374381912571
- Requester VPC: ewb-vpc-a
- Accepter VPC: ewb-vpc-b
- Requester CIDR: 10.10.0.0/16
- Accepter CIDR: 10.20.0.0/16
- Final status: Active

After accepting the peering request, the required routes were added to both private route tables and the public route table used by the bastion.



Active_VPC_Peering



App_A_Security_Group

9. Connectivity Testing and Results

9.1 Baseline Test Before Peering

From App-A, the following command was executed before establishing peering:

ping -c 4 10.20.1.10

Result: Four packets were transmitted, no replies were received, and packet loss was 100%.

This recorded the initial connectivity failure before the peering connection and routes were configured.

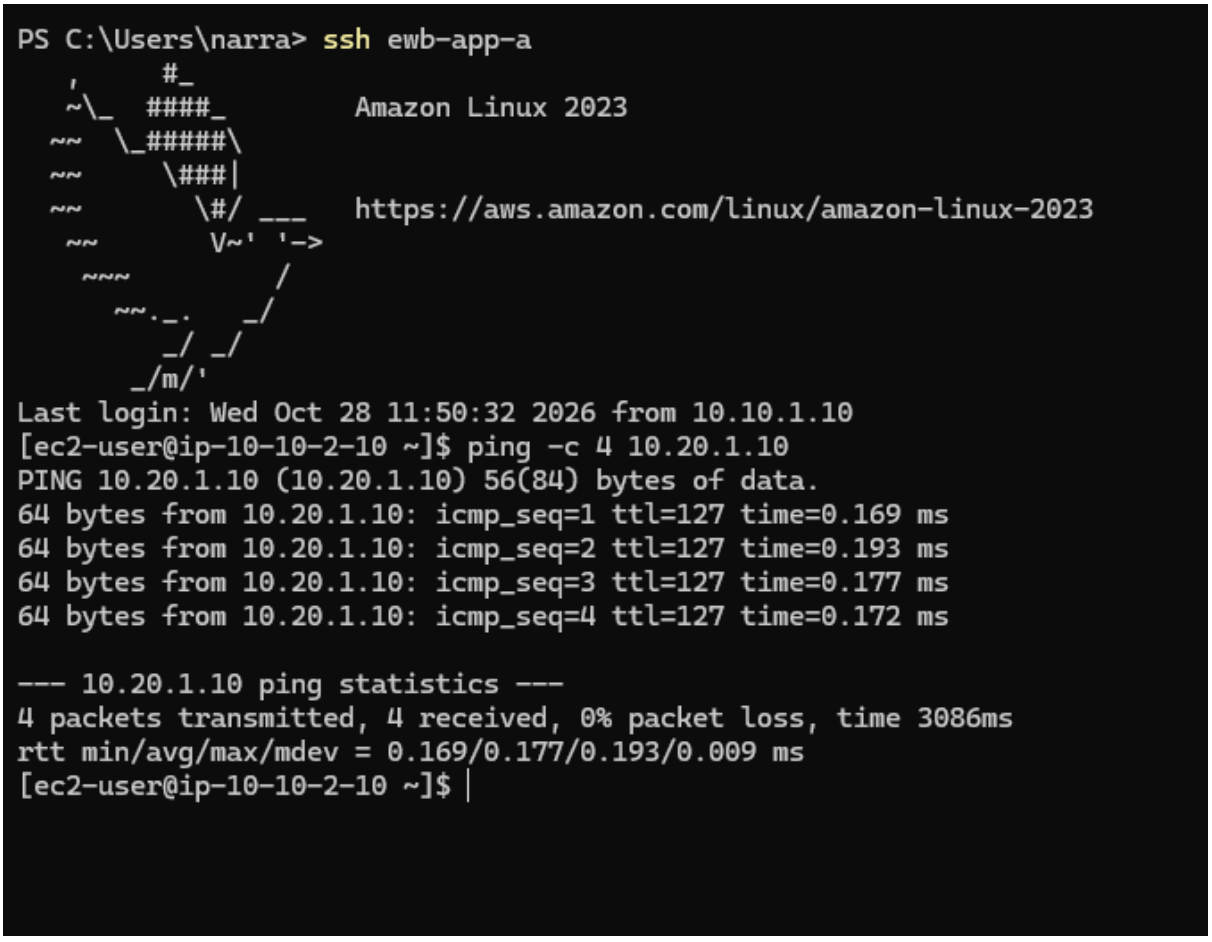
9.2 Forward Connectivity Test: App-A to App-B

After configuring peering and the routes, the same command was executed from App-A:

ping -c 4 10.20.1.10

Result: Four packets transmitted, four received, and 0% packet loss.

This confirms that App-A can reach App-B using the private IP address through the peering connection.



Forward_Ping_Success

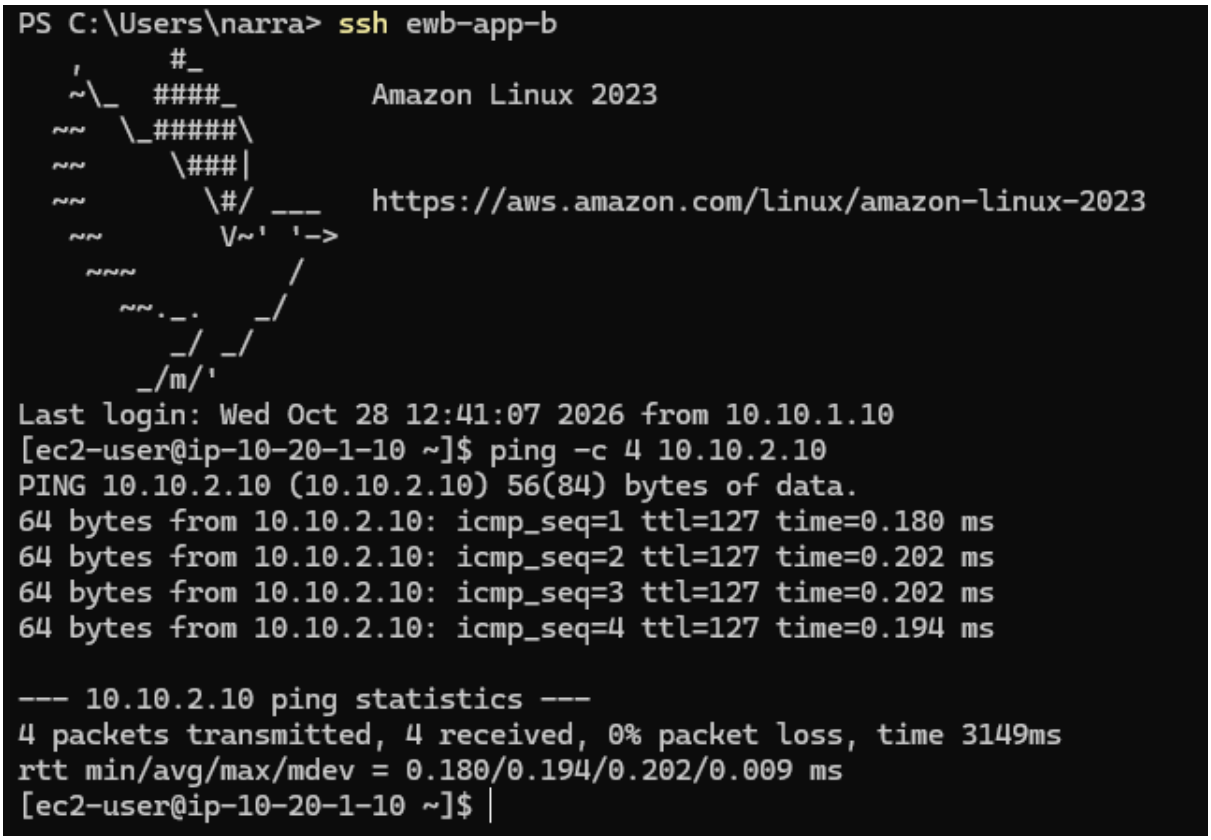
9.3 Reverse Connectivity Test: App-B to App-A

From App-B, the following command was executed:

ping -c 4 10.10.2.10

Result: Four packets transmitted, four received, and 0% packet loss.

This confirms that App-B can reach App-A and verifies bidirectional connectivity.



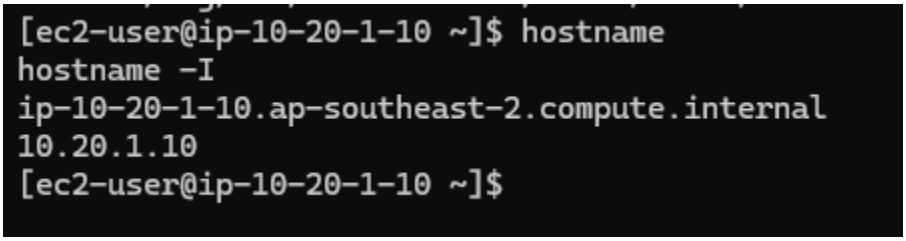
Reverse_Ping_Success

9.4 SSH Access to App-B Through the Bastion

From the local Windows PowerShell terminal, the following command was executed:

ssh ewb-app-b

The SSH configuration used ProxyJump to connect through the bastion. The connection succeeded, and the App-B Amazon Linux terminal was displayed.

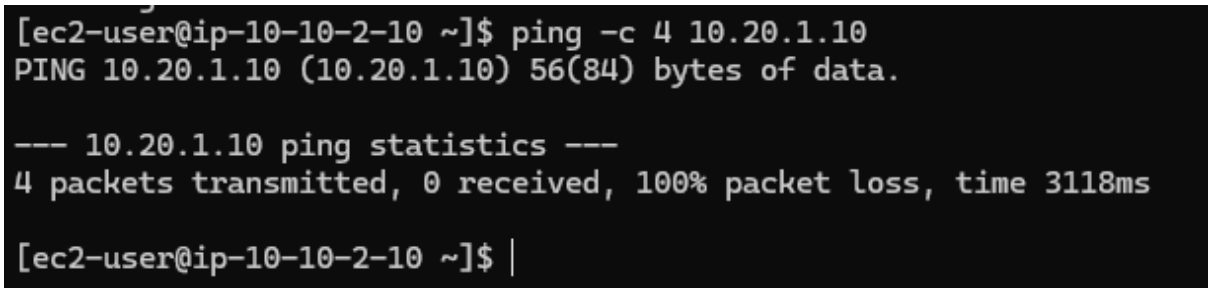


SSH_To_App_B

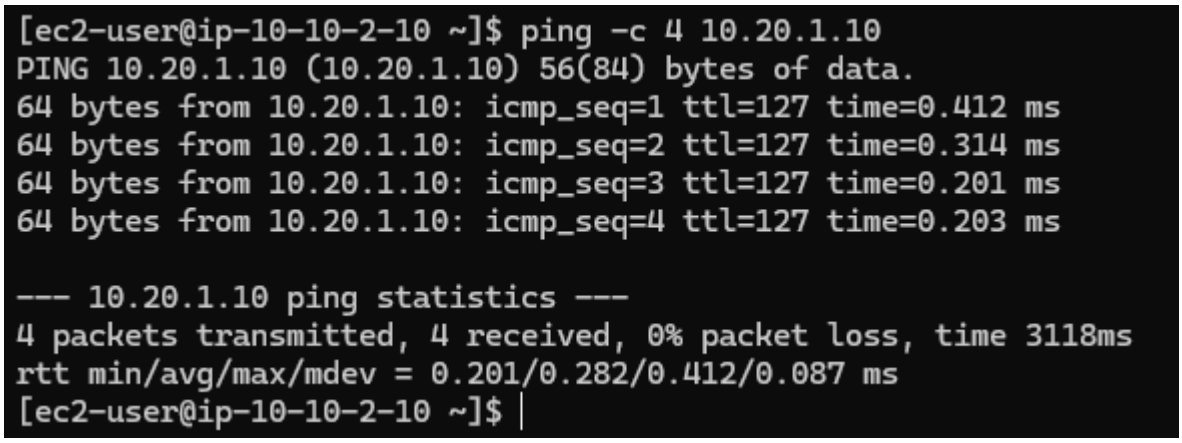
9.5 Controlled ICMP Failure and Recovery

This test is required by the project workbook and must be performed and documented before submission.

1. Temporarily remove only App-B's inbound ICMP Echo Request rule with source 10.10.2.10/32.
2. From App-A, execute ping -c 4 10.20.1.10 and record the failure.
3. Restore the exact ICMP rule.
4. Repeat the ping and record successful recovery.
5. Confirm the final App-B security group contains only the intended SSH and ICMP inbound rules.



Controlled_ICMP_Failure



ICMP_Recovery

10.Conclusion

The project demonstrates how AWS VPC peering enables private communication between two separate VPCs. Correct route table configuration, return routes, security groups and a bastion host are essential to achieving secure connectivity.

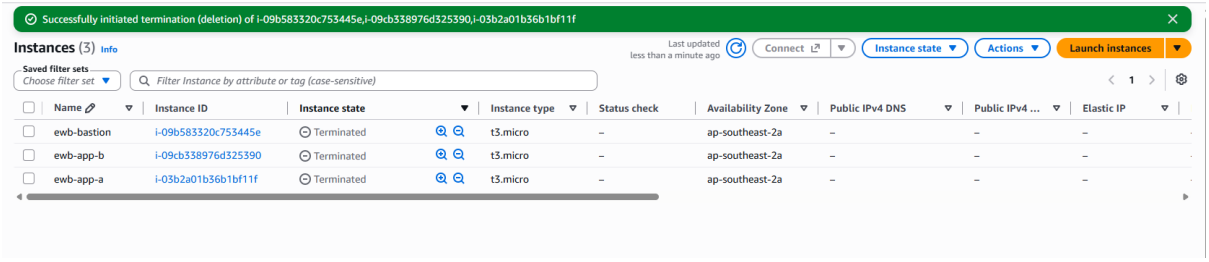
The forward and reverse ping tests succeeded, and SSH access to App-B through the bastion was verified. The missing route in the public route table was identified during troubleshooting and added to allow the bastion to reach App-B.

The controlled ICMP failure and recovery test, along with final cleanup evidence, should be documented before the report is submitted.

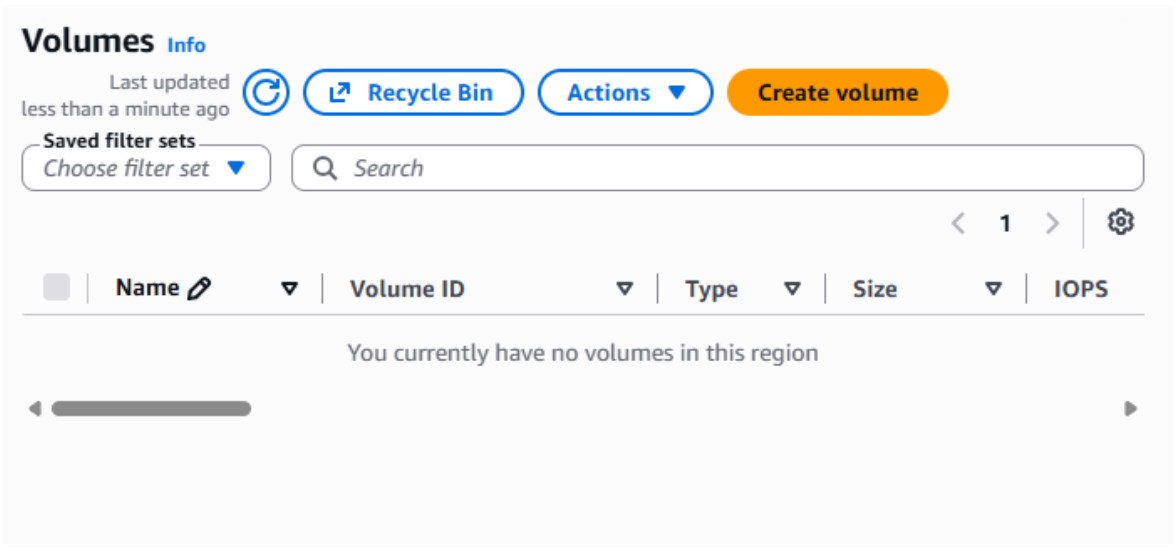
11.Cleanup and Cost Management

After saving all required screenshots and completing the report, follow the workbook's cleanup instructions if the lab resources are no longer required.

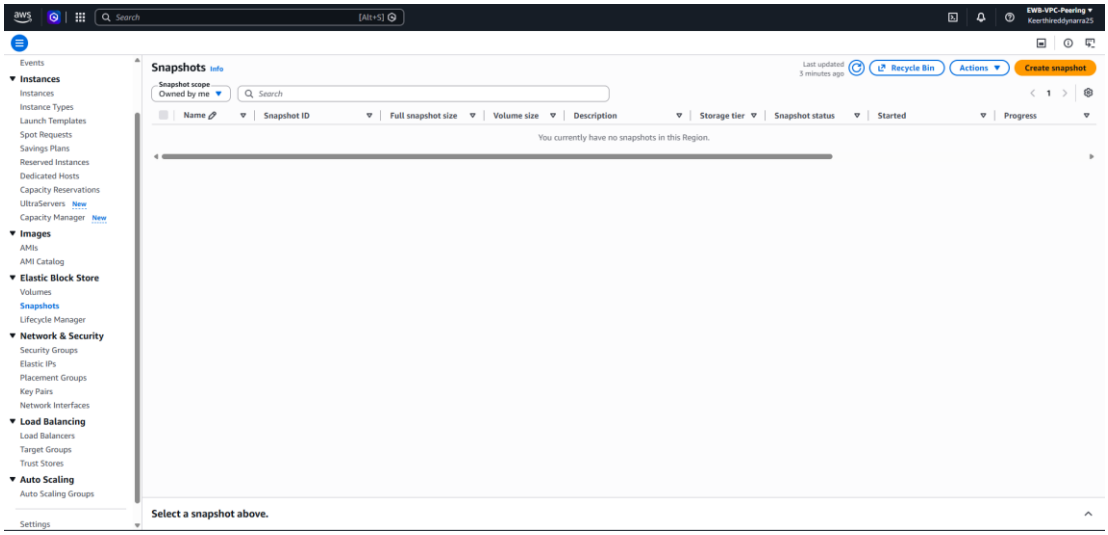
The cleanup should include the three EC2 instances, lab EBS volumes and snapshots, the peering connection, lab security groups, subnets, custom route tables, internet gateway and both VPCs. Remove the lab key pair if it is no longer needed, and verify that no chargeable project resources remain.



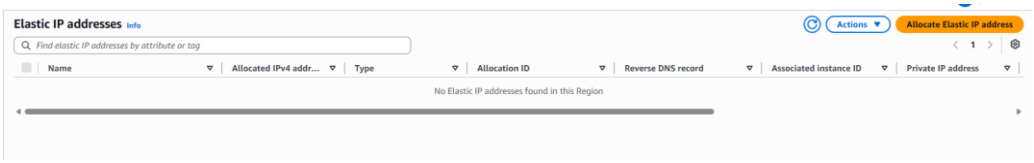
EC2_Instances_Terminated



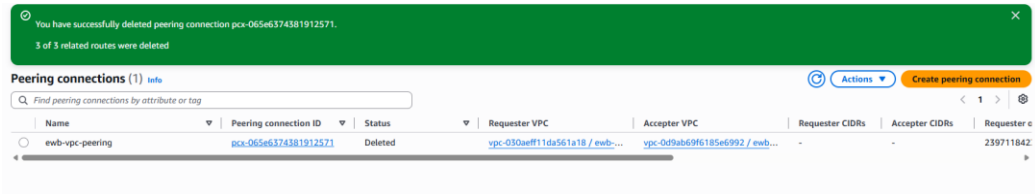
EBS_Volumes_Empty



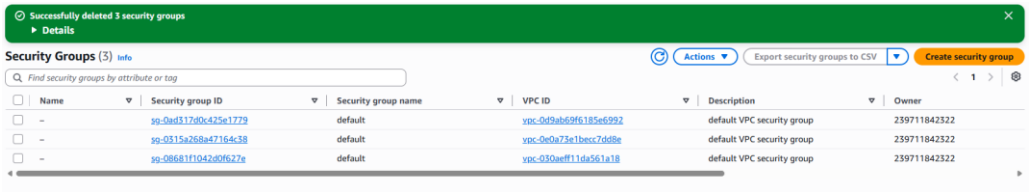
EBS_Snapshots_Empty



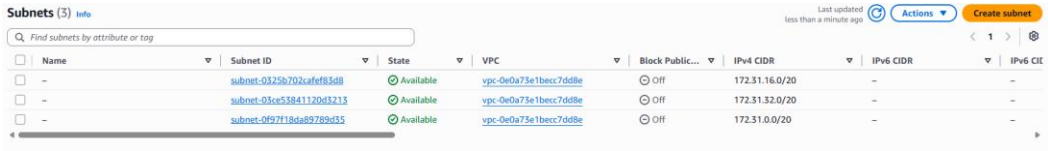
Elastic_IPs_Empty



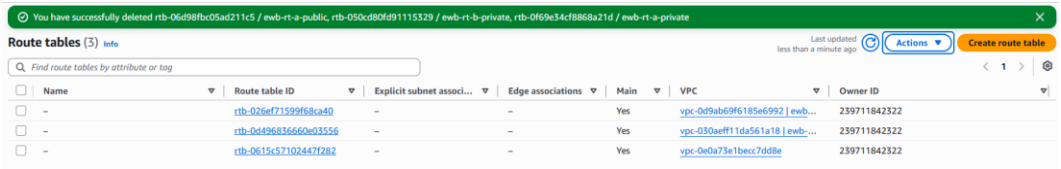
VPC_Peering_Deleted



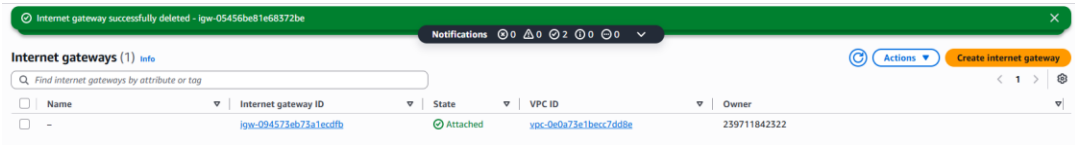
Security_Groups_Deleted



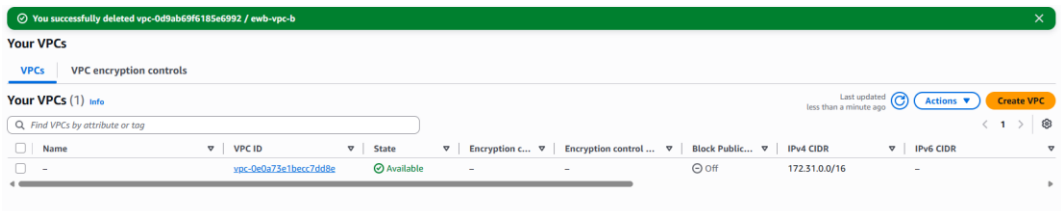
Security_Groups_Deleted



Project_Route_Tables_Deleted



Project_Internet_Gateway_Deleted



Project_VPCs_Deleted

12.Key Learnings

- VPC CIDR blocks must not overlap for peering to work.
- An active peering connection alone does not provide connectivity; route tables and security groups must also permit traffic.
- Both forward and return routes are necessary.
- Private application servers can be administered through a bastion without assigning them public IP addresses.
- Security groups should permit only the required sources and protocols.
- Connectivity should be verified with both successful and controlled failure tests.